YuAng Chen

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Education

The Chinese University of Hong Kong, Shenzhen *2020.01 – 2022.10 (expect)*

• PhD in Computer Science China

• High-Performance Computing for SpMV and Irregular Algorithms, Multicore and Distributed System, Graph Neural Network

EIT Digital Master School *2015.06 – 2018.10*

Entry - Eindhoven University of Technology Netherlands

Exit - Technical University of Berlin Germany

• Dual Master degrees with full EIT Scholarship

• Core Modules: Multicore System, Embedded Visual Control, Embedded Computing Architecture

RWTH Aachen University *2014.10 – 2015.06*

• Exchange undergraduate in Communication Engineering with CSC Scholarship Germany

Huazhong University of Science and Technology *2011.09 – 2014.06*

• Bachelor of Engineering in Electronic Science and Technology China

Publication

YuAng Chen and Yeh-Ching Chung, "Workload Balancing via Graph Reordering on Multicore Systems," *IEEE Transaction on parallel and Distributed Systems* (TPDS), Vol. 33, No. 5, 2022, pp. 1231-1245

YuAng Chen and Yeh-Ching Chung, "HiPa: Hierarchical Partitioning for Fast Page Rank on Multicore Systems," To appear in *Proceedings of IEEE International conference on Parallel Processing* (ICPP), 2021.

YuAng Chen and Yeh-Ching Chung, "Corder: Cache-Aware Reordering For Optimizing Graph Analytics," *Proceedings of ACM International conference on Principles and Practice of Parallel Programming* (PPoPP), 2021

Ongoing:

YuAng Chen and Yeh-Ching Chung, "A Dynamic Caching Strategy for Shared-Memory Graph Analytics", under revision of *IEEE Transactions on Parallel and Distributed Systems*.

YuAng Chen and Yeh-Ching Chung, "Co-Design of Mixed Matrix Format and Processing Engine for Efficient SpMV on Graphs", submitted to *IEEE CLUSTER 2022*

YuAng Chen and Yeh-Ching Chung, " Locality Extraction & Blocking for Graph Adjacency Matrix Multiplications", in preparation for TPDS

Project

GNN in C++ *2020.02 – 2020.04*

• Implement a 2-layer GCN in pure C++ independent of third-party libraries

• Components include Adam Optimizer, ReLU, CrossEntropy; fulfill the mechanisms of forward- and backward-propagation

• Designed for multicore systems; parallelized via OpenMP

Deep GNN *2020.03 – 2020.06*

• Investigate the architecture of deep GNN with application on text classification

• Build an 8-layer GCN with residual connection and k-NN dilation

Academic Experience

Fraunhofer FOKUS *2017.04 – 2018.06*

Research Intern, Germany`s Smart Cities Project Germany

• Worked on the simulation of a SICCT card terminal and its belonging IC cards

• Implemented the TLS secured communication channel between the simulation and the client device

• Deployed the Jenkins CI automated testing for the simulation software

Strength

• Solid skills in Linux, C/C++, Java, Python; Experience with VHDL, Verilog